

Nursing Management of the Child with Fractures



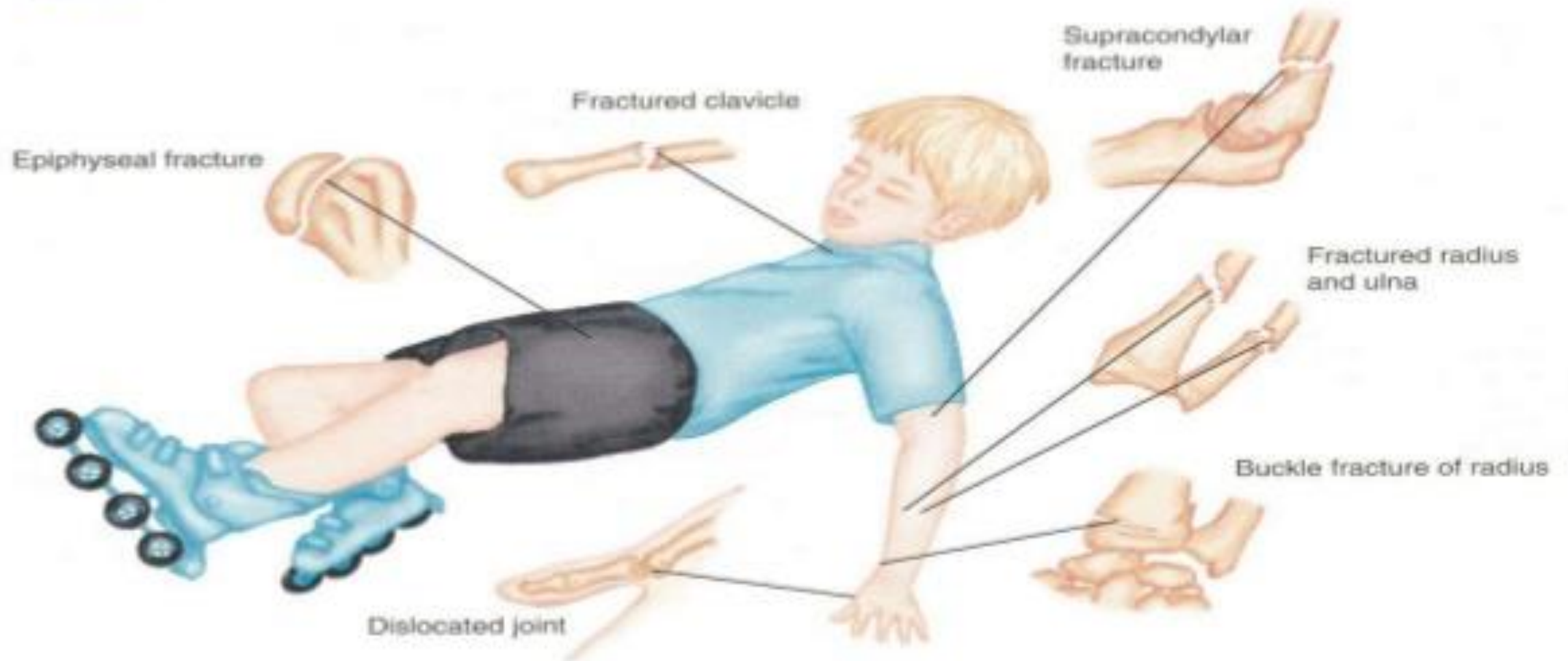
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**UNIT- XV
Musculo Skelital Disorders**

Introduction to Pediatric Fractures

- **Definition:** A break or disruption in the continuity of the bone.
- **Frequency:** Fractures are common in children due to increased mobility and a higher ratio of cartilage to bone.
- **Common Sites:** Distal Forearm (Radius/Ulna) – Most common.
 - Clavicle.
 - Tibia/Fibula.
- **Healing:** Children's bones heal significantly faster than adults due to a thick, active periosteum.

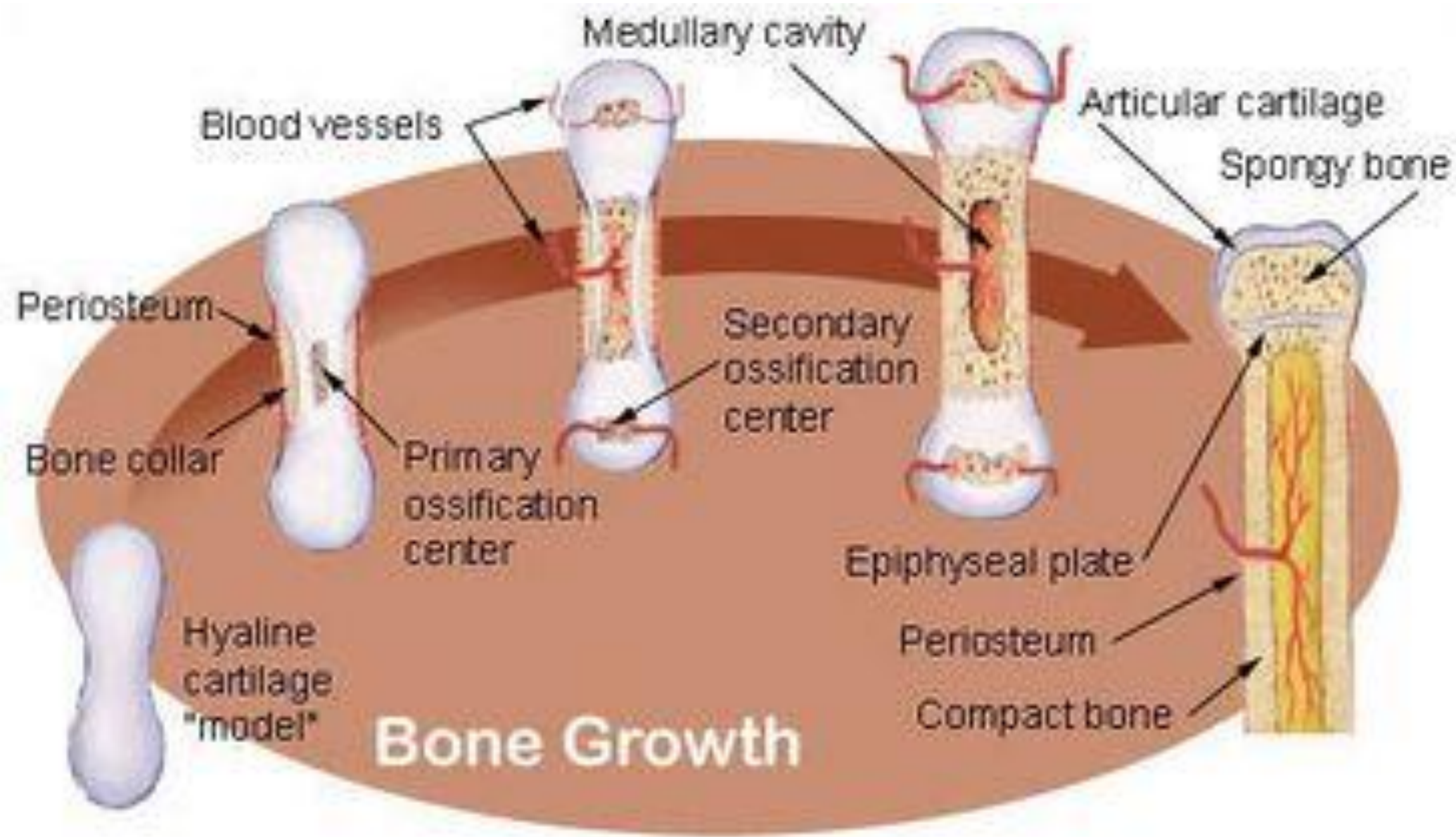
Pediatric Fractures and Falls

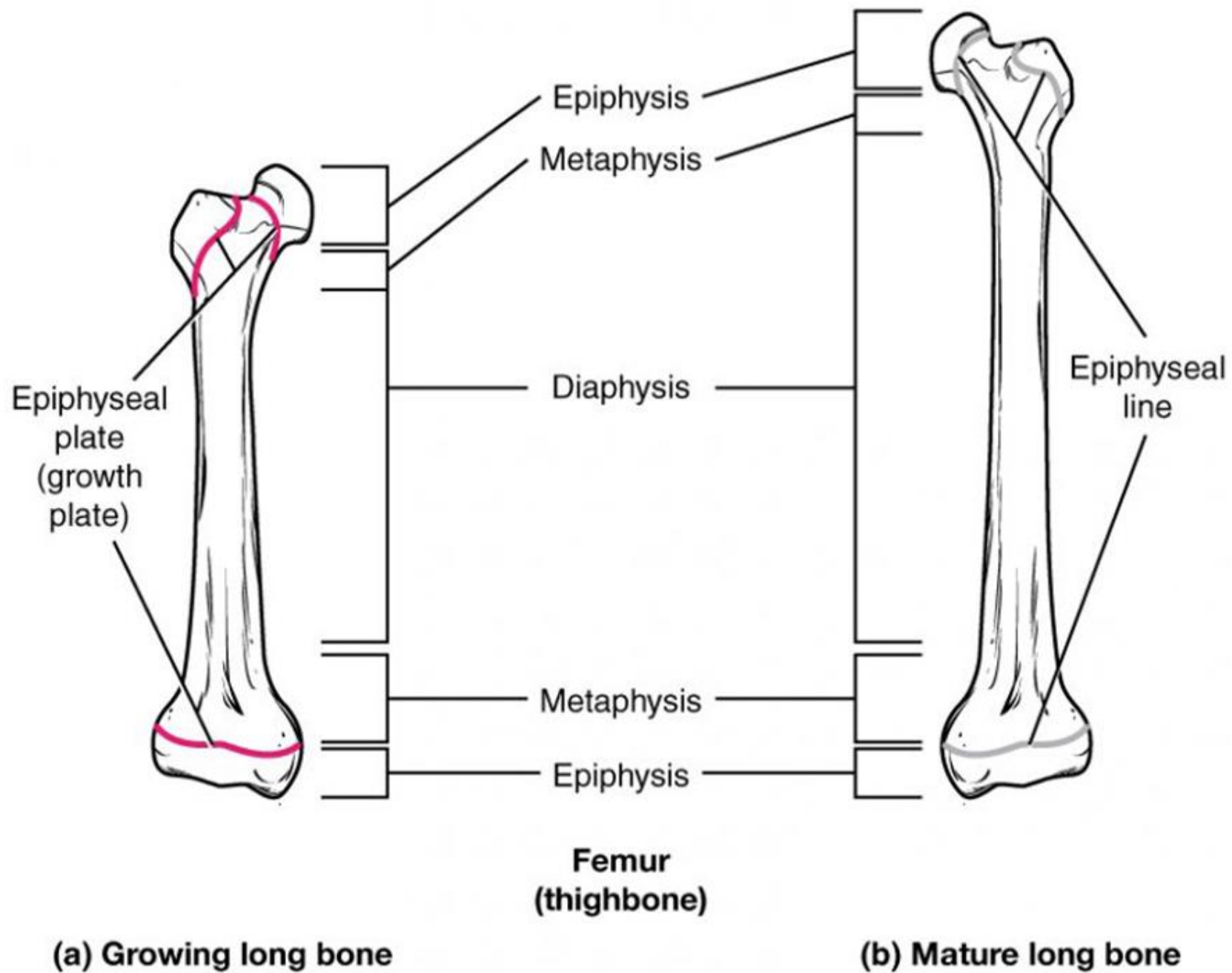


McKinney Text, p. 1424

Pediatric vs. Adult Bone Physiology

- **Porous Bone:** Children's bones are more porous and less dense; they often bend before they break.
- **Periosteum:** Thicker and more vascular in children, promoting faster callus formation.
- **Physis (Growth Plate):** The weakest part of the long bone; injury here can lead to permanent growth disturbances.
- **Remodeling:** Children have a remarkable ability to "straighten" angulated fractures over time through natural growth.

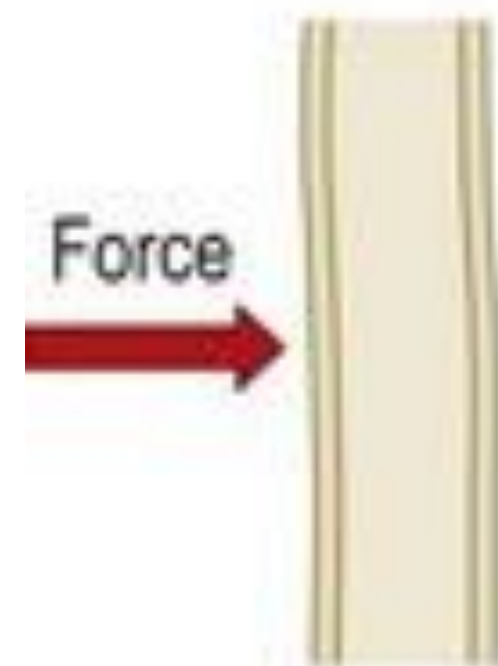




Types of Fractures Unique to Children

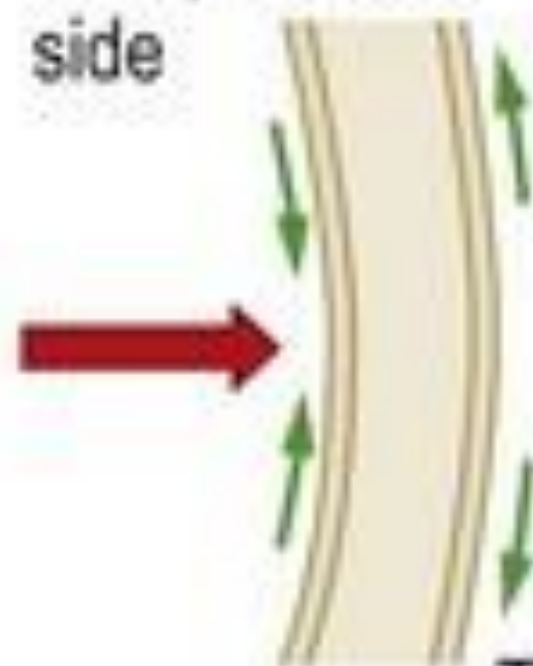
- 1. Plastic Deformation:** Bone is bent but not broken.
- 2. Buckle (Torus) Fracture:** Compression of the porous bone, appearing as a "bulge" on X-ray.
- 3. Greenstick Fracture:** An incomplete break; the bone is broken on one side and bent on the other (like a young green branch).
- 4. Complete Fracture:** Bone is broken into two distinct pieces.

Normal bone



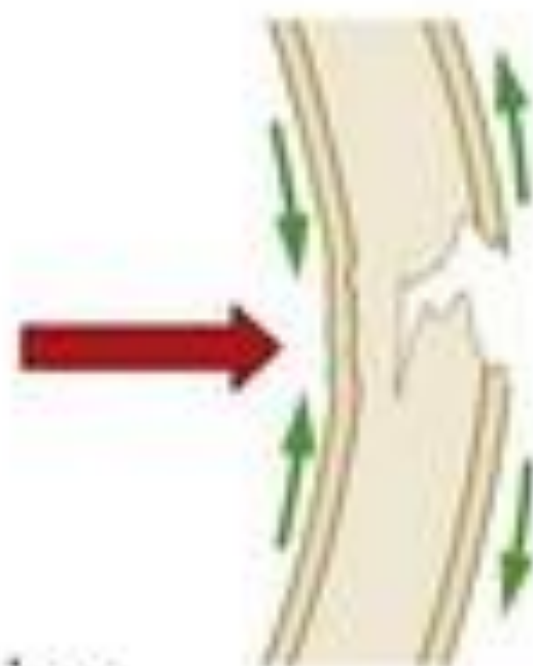
Plastic deformation

Compression side



Tension side

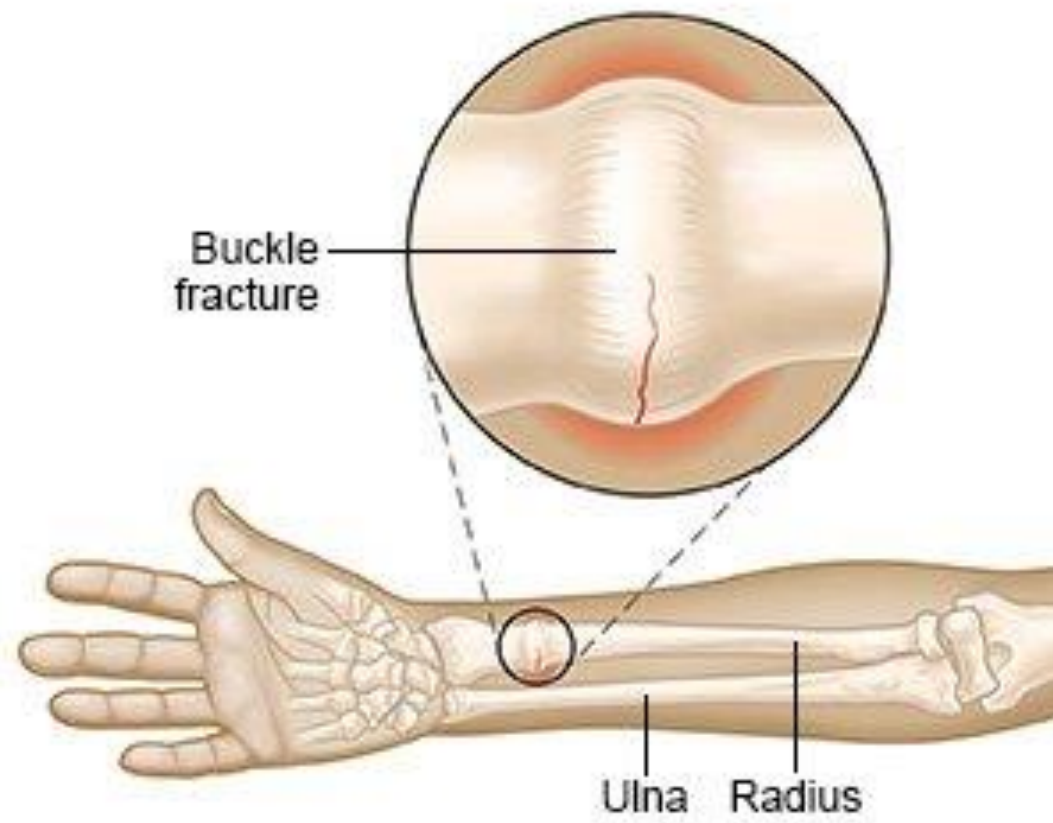
Greenstick fracture



Complete fracture



Buckle Fracture



Growth Plate (Salter-Harris) Injuries

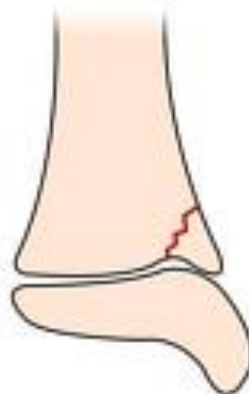
- **Importance:** The epiphyseal plate (growth plate) is responsible for longitudinal growth.
- **Classification:** Salter-Harris system (Types I through V).
- **Nursing Alert:** Any injury involving the growth plate requires long-term follow-up to monitor for limb-length discrepancies or angular deformities.

Salter-Harris classification of physeal fractures

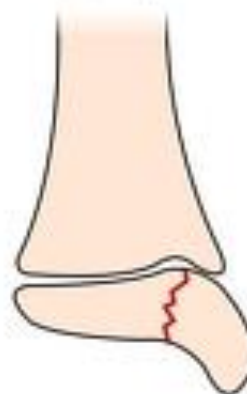
Type I
slipped physis



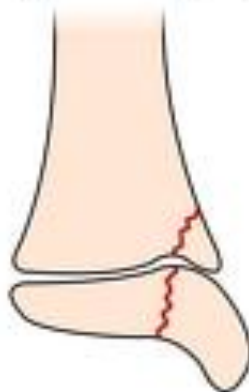
Type II
metaphysis



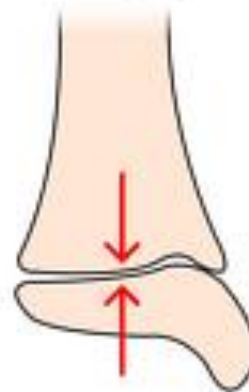
Type III
epiphysis



Type IV
metaphysis + epiphysis



Type V
crushed physis



Clinical Manifestations

- Generalized swelling and bruising.
- Pain or tenderness to touch.
- Deformity or abnormal alignment.
- Diminished functional use (refusal to walk or move a limb).
- Crepitus (grating sensation at the break site).

Nursing Assessment: The "5 Ps"

The Neurovascular Check (Crucial Nursing Task):

- 1.Pain:** Unrelieved by medication or out of proportion to injury.
- 2.Pulselessness:** Distal to the injury (late sign).
- 3.Pallor:** Pale or blue skin, slow capillary refill (>3 seconds).
- 4.Paresthesia:** Numbness, tingling, or "pins and needles."
- 5.Paralysis:** Inability to move digits.

Diagnostic Evaluation

- **X-ray:** The gold standard. Often involves two views (AP and Lateral).
- **Comparison X-rays:** In children, X-rays of the *unaffected* limb may be taken to compare growth plate appearance.
- **History:** Critical to match the mechanism of injury with the fracture type (screening for non-accidental trauma/abuse).

Emergency Nursing Management

- 1.Immobilize:** Splint the joint above and below the fracture site.
- 2.Elevate:** Raise the limb above heart level to reduce edema.
- 3.Cold Application:** Apply ice (intermittently) for the first 24–48 hours.
- 4.NPO Status:** Keep the child NPO in case immediate surgical reduction or sedation is required.

Medical Management: Reduction

- **Closed Reduction:** Realigning the bone manually under sedation or local block (no incision).
- **Open Reduction (ORIF):** Surgical alignment using pins,nails, plates, or screws.
- **Retention:** Once reduced, the bone is held in place by a **Cast, splint, or traction.**

Nursing Diagnosis for Pediatric Fractures

- 1.Acute Pain** related to physical injury and muscle spasms.
- 2.Risk for Peripheral Neurovascular Dysfunction** related to tissue swelling or restrictive cast.
- 3.Impaired Physical Mobility** related to limb immobilization.
- 4.Risk for Impaired Skin Integrity** related to cast edges or bed rest.

Nursing Management of the Child in a Cast

- **Casting Material:** Plaster of Paris (heavy, takes 10-72h to dry) vs. Synthetic/Fiberglass (light, dries in minutes).
- **Handling:** Use **palms, not fingertips**, while the cast is wet to prevent pressure indentations.
- **Positioning:** Elevate on pillows to reduce swelling.
- **Observation:** Check for "hot spots" on the cast or foul odors (signs of infection/pressure sores).

Cast Care Education for Parents

- Keep the cast dry (unless it is a specialized waterproof cast).
- **Do not** stick objects (rulers, knitting needles) inside the cast to scratch itches.
- **Alternative for itching:** Use a blow dryer on a **cool** setting or tap gently on the outside.
- Monitor for "windowing" or skin irritation at the edges.

Traction in Pediatrics

- **Purpose:** To fatigue the muscle and reduce spasms, align bone fragments, and provide immobilization.
- **Types:**
 - **Skin Traction (e.g., Bryant, Buck):** Uses straps/boots applied to the skin.
 - **Skeletal Traction:** Pins or wires are inserted directly into the bone (requires pin site care).

Nursing Care for the Child in Traction

- **Weights:** Must hang freely. Never lift or remove weights unless ordered.
- **Ropes/Pulleys:** Ensure ropes are in the center of the pulley grooves and not frayed.
- **Counter-traction:** Ensure the child's body weight provides enough counter-traction (don't let them slide to the foot of the bed).
- **Skin:** Inspect the heels and sacrum for pressure ulcers.

Complication: Compartment Syndrome

- **Definition:** Increased pressure within a muscle compartment compromising circulation.
- **Nursing Action:** This is a **Surgical Emergency**.
- **Signs:** Severe pain on passive stretch, tense/tight muscle, paresthesia.
- **Treatment:** The nurse must notify the surgeon immediately; a **fasciotomy** may be required.

Complication: Osteomyelitis

- **Definition:** Infection of the bone, common in open fractures or skeletal traction.
- **Signs:** Fever, localized warmth, redness, and refusal to move the limb.
- **Nursing Care:** Administer long-term IV antibiotics as ordered and monitor inflammatory markers (CRP/ESR).

Nutritional Support for Bone Healing

- **Calcium & Vitamin D:** Essential for bone mineralization and callus formation.
- **Protein:** Needed for tissue repair.
- **Fiber & Fluids:** High risk for constipation in children with decreased mobility or those on opioid pain medications.

Discharge Planning & Rehabilitation

- **Weight-Bearing Status:** Ensure parents understand crutch-walking or wheelchair safety.
- **Follow-up:** Schedule X-rays to ensure proper alignment during the healing phase.
- **Cast Removal:** Prepare the child for the loud noise of the cast saw (explain it vibrates and won't cut skin).
- **Skin Care after Removal:** Soaking the limb in warm water to remove dry/dead skin; avoid vigorous scrubbing.

